

The big picture

Guiding question(s)

- What happens when reactants share their electron pairs with others?

Keep the guiding question in mind as you learn the science in this subtopic. You will be ready to answer it at the end of this subtopic. The guiding questions require you to pull together your knowledge and skills from different sections, to see the bigger picture and to build your conceptual understanding.

Have you ever wondered how some foods stay fresh for a long time while other foods spoil quickly? Food spoilage involves a series of chemical reactions and factors such as composition, storage conditions and moisture content determine the rate of food spoilage.

Foods containing certain types of fat have a slow rate of spoilage, allowing for a long shelf-life, as well as the flavours and textures they add to the food (**Figure 1**).



Figure 1. Example of fat-containing food products.

Credit: miguelangelortega, Getty Images

In fact, various aspects of food are interconnected with the chemical reactions occurring within the food. Bond-forming and bond-breaking reactions are essential in the transformation of molecules through various mechanisms, modifying the nutritional composition and sensory characteristics of food.

Prior learning

Before you study this subtopic make sure that you understand the following:

- How to explain the concept of covalent bond (see [section S2.2.1a](#)  ([\(/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43106/\)](https://study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43106/)).
- How to deduce the Lewis formula of molecules and ions with their electron pairs on each atom (see [section S2.2.1a](#)  ([\(/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43106/\)](https://study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43106/)).
- how to deduce the products of reactions of hydrogen with alkenes (see [section R3.2.11](#)  ([\(/study/app/preview-p/sid-426-cid-753301/book/standard-electrode-potentials-hl-id-46335/\)](https://study/app/preview-p/sid-426-cid-753301/book/standard-electrode-potentials-hl-id-46335/)).